DPed-VLN: A Benchmark for Socially Compliant Vision-and-Language Navigation in Dynamic Pedestrian Environments

Supplementary Material

A. Additional Dataset Details

This supplementary document provides additional analyses of DPed-VLN construction, paired instruction conditions, and experimental results, together with implementation details for the reference policies and VLM adaptation.

A.1. Instruction Generation

Figure 1 summarizes the paired instruction generation pipeline, whose VLM component uses Qwen-VL. The level-1 branch uses static scene context and route-level observations for goal-oriented guidance. The level-2 branch additionally receives dynamic-pedestrian context from the recorded trajectory. Both conditions refer to the same navigation episode, and the primary distinction is the pedestrian-related guidance added in level-2. Their comparison evaluates policy responses to route guidance alone and to guidance augmented with pedestrian information.

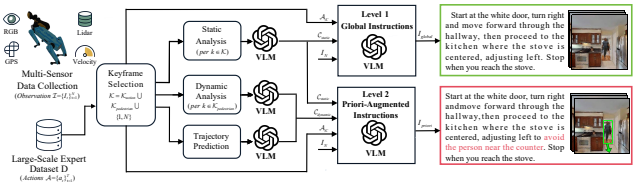


Figure 1. Paired instruction generation. level-1 uses static route context, while level-2 additionally receives pedestrian-related context for diagnostic evaluation.

A.2. Action Distribution

Figure 2 reports the expert action distribution. FORWARD is the largest single category (44.90%); TURNLEFT and TURNRIGHT account for 24.90% and 27.90%. Straight segments account for many navigation steps, while turns and pedestrian avoidance involve action transitions. This distribution motivates the inflection-weighted imitation objective in DPet-IL, which assigns greater weight to such transitions.

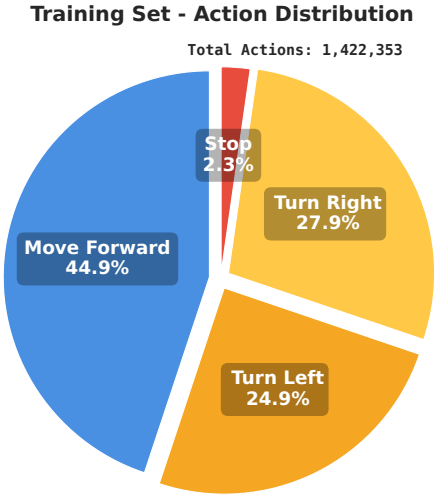


Figure 2. Action distribution in DPed-VLN expert trajectories.

A.3. Instruction Semantics

Figure 3 shows corpus-level lexical differences. level-1 emphasizes static spatial expressions such as rooms, hallways, and directions. level-2 retains navigation language while incorporating terms related to people, avoidance, and spatial interaction, reflecting the high-level pedestrian priors injected by the two-stage generation pipeline.

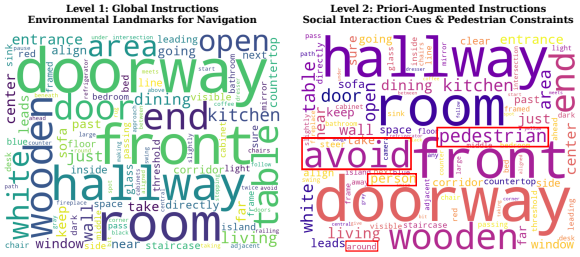


Figure 3. Stopword-filtered word clouds for level-1 and level-2 instructions.

Figure 4 quantifies pedestrian-related vocabulary. Pedestrian and avoidance terms occur substantially more frequently in level-2, consistent with the additional dynamic-pedestrian context provided under this condition.

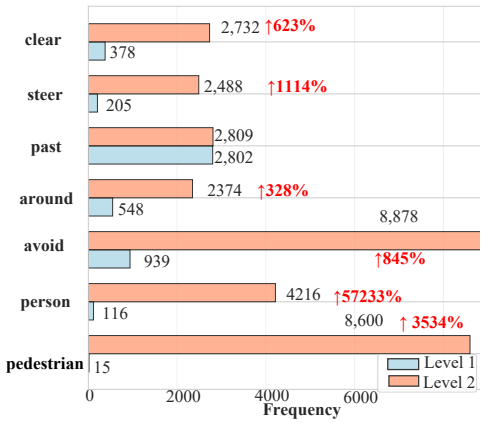


Figure 4. Frequency comparison of pedestrian-related keywords in level-1 and level-2 instructions.

A.4. Instruction Length Distributions

Figure 5 compares word-count distributions across *train*, *val-unseen*, and the two instruction conditions. The split distributions are closely aligned across both levels, confirming consistency between the training and unseen evaluation corpora.

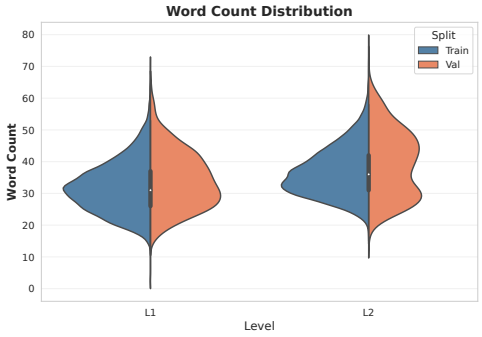


Figure 5. Instruction word-count distributions across *train* and *val-unseen* splits and instruction levels.